

BIELSKO-BIALA, Poland

WMO: 126000

Lat: 49.8067N

Lon: 19.0023E

Elev: 1311

StdP: 14.01

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 6.4	(c) 11.1	(d) 1.4	(e) 6.2	(f) 7.6	(g) 6.4	(h) 8.0	(i) 13.6	(j) 34.1	(k) 41.7	(l) 29.6	(m) 38.5	(n) 5.2	(o) 60	(p) 0.532

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 16.5	(c) 84.9	(d) 67.7	(e) 81.4	(f) 66.4	(g) 78.4	(h) 64.9	(i) 69.6	(j) 80.7	(k) 67.9	(l) 78.5	(m) 66.3	(n) 76.1	(o) 6.6	(p) 190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 65.8	(b) 99.8	(c) 74.9	(d) 64.1	(e) 94.0	(f) 72.8	(g) 62.6	(h) 89.0	(i) 70.6	(j) 34.4	(k) 80.8	(l) 32.9	(m) 78.6	(n) 31.6	(o) 76.1	(p) 80.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 25.7	(b) 21.1	(c) 17.6	(e) 1.8	(f) 89.9	(g) 6.8	(h) 2.8	(i) -3.1	(j) 91.9	(k) -7.1	(l) 93.6	(m) -10.9	(n) 95.2	(o) -15.9	(p) 97.3		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			1.2	73.7	6.7	2.6	-3.7	75.6	-7.6	77.1	-11.4	78.6	-16.3	80.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.2	29.6	32.2	38.2	48.4	56.1	62.2	65.6	65.1	57.0	49.1	41.1	32.6	
	DBStd	15.04	10.06	9.95	8.52	8.43	7.00	6.66	6.12	6.08	6.69	8.17	9.05	9.43	
	HDD50	2617	633	499	373	125	22	1	0	0	14	119	290	541	
	HDD65	6371	1098	918	831	500	284	132	68	73	252	493	717	1005	
	CDD50	1958	0	1	8	77	213	366	483	470	223	92	23	2	
	CDD65	235	0	0	0	2	10	47	86	78	11	1	0	0	
	CDH74	1873	0	0	0	17	89	377	702	613	72	3	0	0	
	CDH80	444	0	0	0	1	7	84	183	157	12	0	0	0	
Wind	WSAvg	6.9	8.2	8.1	7.9	6.9	6.0	5.6	5.7	5.3	6.0	7.0	7.7	8.4	
Precipitation	PrecAvg	36.6	2.0	1.9	2.3	2.5	4.3	4.6	5.1	3.5	3.6	2.5	2.4	1.9	
	PrecMax	53.8	4.3	4.3	5.5	5.6	17.0	9.6	13.7	7.9	8.5	6.4	4.3	4.8	
	PrecMin	27.9	0.7	0.6	0.8	0.0	1.3	1.6	0.7	0.8	1.2	0.5	0.0	0.5	
	PrecStd	5.9	0.8	0.9	1.1	1.2	2.9	2.0	2.8	1.6	1.9	1.5	1.0	0.8	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.2	56.5	63.6	76.4	80.4	86.9	89.2	89.7	81.0	72.2	64.2	56.2	
		MCWB	45.9	45.6	49.8	58.3	62.2	68.5	68.2	68.3	63.1	59.7	53.6	47.9	
	2%	DB	49.5	52.4	58.7	70.7	76.4	82.4	85.1	84.0	75.2	68.4	60.5	52.0	
		MCWB	43.9	44.3	47.3	55.0	61.5	67.4	68.1	67.4	62.1	58.1	51.8	45.6	
	5%	DB	46.4	49.1	55.0	66.0	73.0	78.5	81.2	80.4	72.0	65.1	57.6	48.3	
		MCWB	41.5	42.6	45.8	53.0	60.2	65.2	66.6	66.6	60.9	56.5	50.0	42.8	
	10%	DB	43.1	45.5	51.0	61.9	69.2	74.7	77.8	76.9	68.5	61.8	54.3	45.3	
		MCWB	39.2	40.4	43.8	50.8	58.3	63.2	65.0	64.8	59.4	54.6	48.3	41.1	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.9	47.4	50.8	59.5	65.7	71.2	71.9	71.6	66.3	61.6	54.9	48.6
			MCDB	51.6	53.8	60.5	72.4	75.9	82.4	83.3	81.8	76.3	69.2	61.9	54.7
2%		WB	44.3	45.2	48.5	56.4	63.1	68.7	69.5	69.1	63.8	59.2	52.5	46.2	
		MCDB	49.0	51.1	56.6	68.4	73.1	79.1	80.6	80.5	72.6	66.3	59.5	51.5	
5%		WB	42.0	43.0	46.5	54.1	61.2	66.4	67.9	67.4	62.0	57.2	50.6	43.4	
		MCDB	46.2	48.4	53.5	64.0	70.9	75.9	78.5	78.3	69.6	63.8	56.9	47.4	
10%		WB	39.5	40.7	44.4	51.8	59.2	64.4	66.2	65.7	60.2	55.0	48.5	41.2	
		MCDB	42.8	45.4	50.4	60.6	68.0	73.1	75.8	75.4	67.1	60.9	53.6	44.9	
Mean Daily Temperature Range	5% DB	MDBR	9.1	10.1	12.2	15.8	16.4	16.0	16.5	17.1	14.2	12.6	10.3	9.3	
		MCDBR	11.3	14.7	18.6	21.9	22.3	21.8	22.5	21.9	19.6	17.3	14.1	11.7	
		MCWBR	8.1	9.5	10.8	11.2	11.4	10.5	9.8	9.8	9.9	10.0	8.6	7.9	
	5% WB	MCDBR	10.8	13.6	16.5	19.8	20.3	19.5	19.8	19.8	17.2	16.4	12.8	10.7	
		MCWBR	8.1	9.5	10.4	10.7	11.4	10.5	10.1	10.0	9.7	10.1	8.2	7.9	
Clear-Sky Solar Irradiance	taub	0.305	0.330	0.373	0.425	0.422	0.424	0.437	0.429	0.398	0.365	0.336	0.303		
	taud	2.301	2.271	2.233	2.162	2.218	2.246	2.216	2.244	2.302	2.373	2.386	2.367		
	Ebn at Noon	231	253	261	259	266	266	259	255	251	239	218	216		
	Edh at Noon	22	29	36	43	43	42	43	40	34	26	21	18		
All-Sky Solar Radiation	RadAvg	275	508	856	1276	1533	1634	1580	1428	1010	612	308	216		
	RadStd	48	81	138	157	196	170	192	130	152	89	45	26		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+1.06	N/A	N/A	+1.93	N/A	N/A	N/A	N/A	N/A	+76
(47) Regional (42 neighbors)		+0.97	N/A	N/A	+1.48	+0.72	+0.61	-229	-302	+135	+54

Nomenclature: See separate page